Chapter 2
Factor Models and Dimensionality Reduction
Appendix 2 - Firm Characteristics in Asset Pricing

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Roadmap

The Problem with Static Betas — A Concrete Example

Consider two firms at the **same point in time**:

small, distressed, loser	*large, growth, winner*
Firm A	**Firm B**
Market cap: \$50M	Market cap: \$80B
B/M ratio: 1.8	B/M ratio: 0.2
Past 12m return: -30%	Past 12m return: $+45\%$
Leverage: 0.8	Leverage: 0.1

Key question: should Firm A and Firm B have the *same* market beta? The *same* sensitivity to a recession shock?

The static model assumption

In static factor models (FF, PCA), β_i is **constant** for each firm. Both firms get a single number estimated from 60 months of past returns. This ignores all the information in their current observable attributes.

Characteristics as Signals of Risk Exposure

Economic theory and empirical evidence suggest that **observable firm attributes predict how a firm responds to systematic shocks**:

Characteristic	**Economic rationale**	**Expected effect on β**
Size (market cap)	Small firms more exposed to liquidity and credit shocks	Small $\Rightarrow$ higher distress β
Value (B/M ratio)	High B/M firms have more assets in place, less flexibility	High B/M $\Rightarrow$ higher value β
Momentum (past returns)	Recent winners differ in investor attention, short-sale pressure	High mom $\Rightarrow$ different factor loading
Leverage	More debt amplifies equity sensitivity to macro shocks	High lev $\Rightarrow$ higher market β
Profitability	Profitable firms less exposed to financial distress	High prof $\Rightarrow$ lower crash β

Characteristics as Signals of Risk Exposure

The key insight

Characteristics are **proxies for latent risk exposures**. They contain information about β that is not captured by past returns alone.

Definition: Firm Characteristics

Definition

A **firm characteristic** $z_{i,p,t-1}$ is any observable attribute of firm i, measured at time $t-1$, that carries information about its future returns or risk exposures.

Characteristics are collected into a vector for each firm and period:

$$z_{i,t-1} = \begin{pmatrix} z_{i,1,t-1} \\ z_{i,2,t-1} \\ \vdots \\ z_{i,P,t-1} \end{pmatrix} \in \mathbb{R}^P$$

They come from **different data sources**, updated at different frequencies:

Source	Frequency	Examples
Financial statements	Annual / Quarterly	B/M, leverage, profitability, investment
Market data	Monthly	Size, momentum, volatility, liquidity

Categories of Characteristics

Price & Trend mom1m, mom6m, mom12m, maxret, chmom	**Value** bm, ep, cfp, dp, sp	**Profitability** roaq, roic, gma, operprof, roavol	**Investment** agr, invest, egr, chcsho, hire
Risk beta, betasq, retvol, idiovol, stdacc	**Liquidity** turn, std_turn, mvel1, dolvol, ill, zerotrade	**Financing** acc, chfeps, baspread, depr, pchgm	**Industry** SIC dummies 14 indicators

Preprocessing: Rank Normalization

Raw characteristics are **not** used directly. They are preprocessed to ensure comparability across firms and over time.

Procedure for each characteristic p at each date t:

1. **Cross-sectional rank:** for each date t, rank all N_t firms by characteristic p

$$\text{rank}_{i,p,t} \in \{1, 2, \ldots, N_t\}$$

2. **Normalize** to the interval $(-1, +1)$:

$$\tilde{z}_{i,p,t} = 2 \cdot \frac{\text{rank}_{i,p,t} - 1}{N_t - 1} - 1$$

Preprocessing: Rank Normalization

Why rank normalization?

- Eliminates **outliers** (e.g. extreme P/E ratios)
- Makes all characteristics **comparable** in scale
- Removes distributional changes over time
- Standard in empirical asset pricing ML

Example

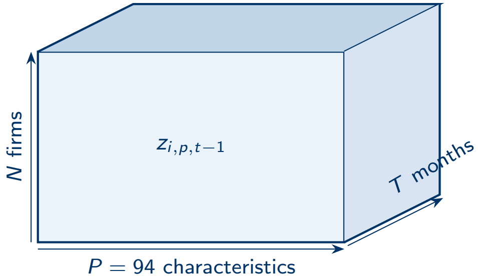
If there are 3000 firms and Apple ranks 2700th by size, its normalized size characteristic is:

$$\tilde{z} = 2 \cdot \frac{2699}{2999} - 1 \approx +0.80$$

A micro-cap ranked 50th would get ≈ -0.97.

The Characteristics Panel: Data Structure

For N firms observed over T months, each with P characteristics, the full dataset is a **three-dimensional panel**:



The Characteristics Panel: Data Structure

The characteristics matrix at time t

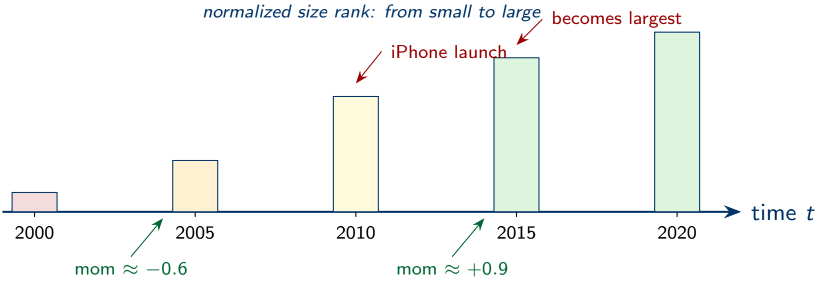
For a given month t, we stack the characteristic vectors of all N_t active firms:

$$Z_{t-1} = \begin{pmatrix} z'_{1,t-1} \\ z'_{2,t-1} \\ \vdots \\ z'_{N_t,t-1} \end{pmatrix} \in \mathbb{R}^{N_t \times P}$$

This matrix changes every month as firms enter and exit the sample.

Time Variation: Characteristics Change Every Month

Apple Inc. (AAPL) — selected characteristics over time (illustrative):



The implication for betas

As Apple grew from a small struggling firm to a mega-cap growth company, its exposure to systematic risk factors changed **continuously**. A static beta estimated over a 5-year window cannot capture this evolution. Characteristics provide a **real-time signal** about current risk exposure.

The Conditional Factor Model

We now have all the ingredients to write the **general conditional factor model**:

General Conditional Factor Model (Eq. 1 of the paper)

$$r_{i,t} = \beta(z_{i,t-1})' f_t + u_{i,t}$$

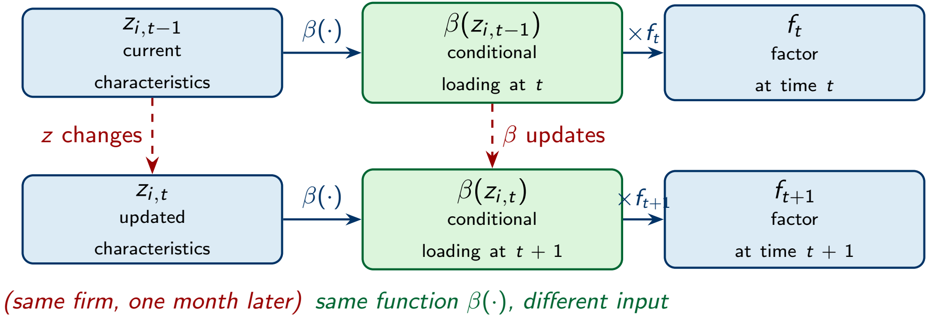
Symbol	Dimension	Meaning
$r_{i,t}$	scalar	excess return of stock i at time t
$z_{i,t-1}$	$P \times 1$	**characteristics** of firm i known at $t-1$
$\beta(z_{i,t-1})$	$K \times 1$	**conditional** factor loadings — a function of z
f_t	$K \times 1$	latent factors at time t
$u_{i,t}$	scalar	idiosyncratic return

The Conditional Factor Model

What changed relative to the static model?

In the static model: β_i is a fixed vector estimated once per firm.
Here: $\beta(z_{i,t-1})$ is a **function** evaluated freshly every month using the firm's current observable attributes.

How Characteristics Generate Time-Varying Betas



Key structural assumption: the mapping $\beta(\cdot)$ is the **same function for all firms at all times**. Time variation in betas comes entirely from time variation in characteristics.

Cross-Sectional and Time-Series Variation — Simultaneously

The conditional model captures **two sources of variation** with one object:

Cross-sectional variation (same t)

At any given month, two firms with **different** $z_{i,t-1}$ get **different** betas:

$$z_{A,t-1} \neq z_{B,t-1} \Rightarrow \beta_A \neq \beta_B$$

Example: a small-cap and a large-cap in the same month have different market beta exposures.

Time-series variation (same firm)

For a single firm, betas change over time as its characteristics evolve:

$$z_{i,t-1} \neq z_{i,s-1} \Rightarrow \beta_{i,t} \neq \beta_{i,s}$$

Example: the same firm has a different beta when it is financially distressed vs. when it is healthy.

Cross-Sectional and Time-Series Variation — Simultaneously

Contrast with static models

	Cross-sectional	**Time-series**
Static (FF, PCA)	✓ (different β_i per firm)	× (constant)
Conditional	✓	✓

The Open Question: Choosing $\beta(\cdot)$

The conditional factor model $r_{i,t} = \beta(z_{i,t-1})' f_t + u_{i,t}$ requires specifying the function $\beta : \mathbb{R}^P \to \mathbb{R}^K$.

This is a **non-parametric estimation problem**: we have n inputs and need to learn a function from data.

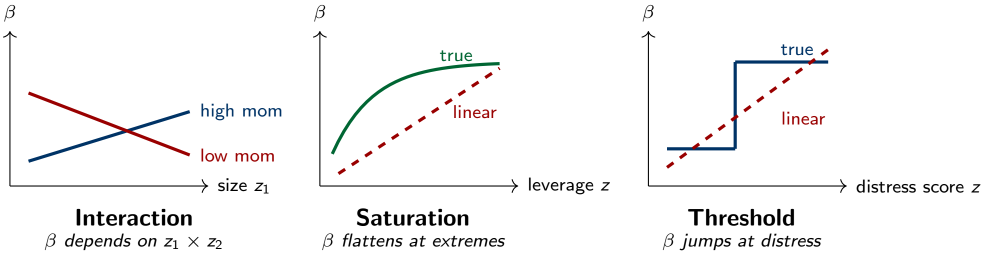
Approach	Specification	Comment
IPCA	$\beta(z)' = z'\Gamma, \quad \Gamma \in \mathbb{R}^{P\times K}$	Simple, closed-form; misses nonlinear interactions
CA0	Single linear layer	Equivalent to IPCA
CA1	1 hidden layer (ReLU)	Captures pairwise interactions
CA2	2 hidden layers (ReLU)	Richer nonlinearities
CA3	3 hidden layers (ReLU)	Maximum flexibility

The Open Question: Choosing $\beta(\cdot)$

Why nonlinearity?

Leading structural models (Campbell–Cochrane 1999, Bansal–Yaron 2004, He–Krishnamurthy 2013) all predict that the mapping from state variables to betas is **inherently nonlinear**. Linearity is a computational convenience, not a theoretical prediction.

Nonlinear Effects That IPCA Cannot Capture



IPCA assumes linearity in all three panels

The dashed lines show what IPCA fits. The solid curves show what the true relationship might look like. Neural networks can approximate all three patterns.

Summary: The Role of Characteristics

Observable attributes $z_{i,t-1}$
size, value, momentum, leverage,
profitability, . . . (94 total)

function $\beta(\cdot)$

Conditional betas $\beta(z_{i,t-1})$
risk exposures updated every month

$\times$ latent factors f_t

Return $r_{i,t} = \beta(z_{i,t-1})' f_t + u_{i,t}$
explained by fac-
tor exposure + noise

Summary: The Role of Characteristics

What characteristics provide:

- Real-time, firm-specific risk information
- Cross-sectional differentiation
- Time-series updating without re-estimation
- A high-dimensional input for the beta network

The remaining question:
What is the function $\beta(\cdot)$?

- IPCA: linear ($\beta = \Gamma' z$) — simple but restrictive
- Conditional Autoencoder: neural network — flexible, nonlinear
- Theory predicts nonlinearity; data confirms it